7. Answer : option(c)

$Ti^{4+}\left(d^0 \Rightarrow 0BM\right)$

$Co^{2+}\left(d^7 \text{spin free} \Rightarrow t_{2g}{}^5, e_g{}^2\ ;\ n=3\ ;\ \mu=3.9BM\right)$

$Cu^{2+}\left(d^9 \text{Low spin} \Rightarrow t_{2g}{}^6, e_g{}^3\ ;\ n=1\ ;\ \mu=1.732BM\right)$

$Ni^{2+}\left(d^8 \text{Low spin} \Rightarrow t_{2g}{}^6, e_g{}^2\ ;\ n=2\ ;\ \mu=2.44BM\right)$

8. Answer : option(b)

The electronic configuration $t_{2g}{}^3, e_g{}^2$

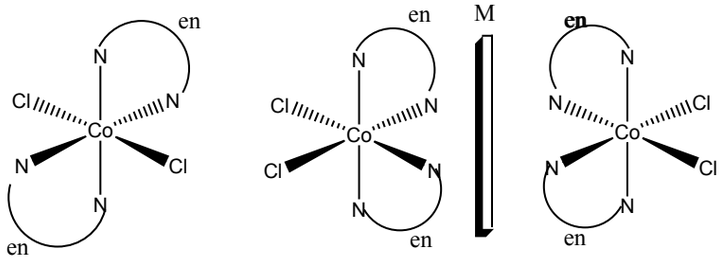
$\left[3\times(-0.4)+2(0.6)\right]\Delta_0$

$\left[-1.2+1.2\right]\Delta_0 = 0$

9. Answer : option(a)

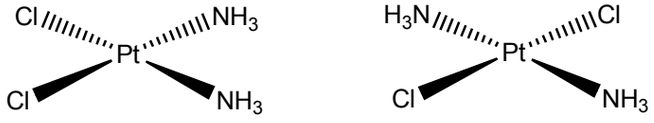
In all the complexes, the central metal ion is Co^{3+}, among the given ligands CN^- is the strongest ligand, which causes large crystal field splitting i.e maximum Δ_0

10. Answer : option(b)



Complexes given in other options (a), (c) and (d) have symmetry elements and hence they are optically inactive.

11. Answer : option(d)



12. Three isomers. If we consider any one of the ligands as reference (say Py), the arrangement of other three ligands (NH_3, Br^- and Cl^-) with respect to (Py) gives three geometrical isomers.

13. Answer : option(c)

(a)coordination isomers

(b) no isomerism (different molecular formula)

(c) $\leftarrow NCS$, $\leftarrow SCN$ coordinating atom differs : linkage isomers

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